Lesson 37: Final Project: Your Research Proposal in APA Style

English Stage 4 (Psychology) · Unit 5: Writing and Presenting Research in APA Style

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 2.9 Overview of the Scientific Method (Summary); 11.5 Presenting Your Research (Summary).

In this lesson

This lesson brings the stage together: you review the research cycle and the key ideas of the course, then assemble, revise and check your complete research proposal in APA style. In English, you practise precise definitions and contrasts, careful claims about evidence, and proofreading.

Learning goals

After this lesson you should be able to:

- Describe the research cycle and place your own proposal in it.
- Define and contrast key ideas: theory and hypothesis, internal and external validity, Type I and Type II errors.
- Assemble a complete proposal in the correct APA order and format.
- Proofread a text at all three levels of APA style.
- Write an abstract that summarizes the whole proposal.

Key language

Term or phrase	Meaning	Example
research literature	all the published research in a particular field	journal articles and scholarly books
interestingness	a property of a research question: the answer is in doubt, fills a gap in the literature, has important practical implications	Is the answer in doubt?
feasibility	the extent to which a research question can be answered with available resources	time, money, skills, equipment, participants
theory	a coherent explanation or interpretation of one or more phenomena	broad: explains larger bodies of data
hypothesis	a prediction about a new phenomenon that would be observed if a particular theory were true	Students who nap after learning will recall more word pairs.

Term or phrase	Meaning	Example
hypothetico-deductive method	the general way researchers use theories to generate new research and, in the process, test and revise the theories	theory, hypothesis, test, reevaluation
internal validity	how far the design supports the conclusion that the independent variable caused the differences in the dependent variable	high in laboratory experiments
external validity	how far the results can be generalized to people and situations beyond those studied	often higher in field studies
Type I error	concluding that an effect is real when it is not	"The nap helps" when it does not
Type II error	failing to reject the null hypothesis when it is false: concluding there is no effect when there is one	missing a real effect of the nap
that is	introduces a restatement or definition	"statistically significant, that is, ... unlikely to be due to chance alone"
in light of	considering	"reevaluate the theories in light of the new results"

Rules and explanations

The ideas

The research cycle (2.9). "A research question based on the research literature leads to an empirical study, the results of which are published and become part of the research literature." Your proposal covers the first steps; Lessons 33 to 36 showed the last one.

What the two summaries say, and where it goes in your proposal

Key idea	Where it appears
Review the literature early: to refine the question, find methods, place the question in context, prepare to write	literature review; references
Interestingness (answer in doubt, a gap, practical implications) and feasibility (time, money, knowledge and skill, equipment, participants)	opening; method
A theory is broad and explains larger bodies of data; a hypothesis is specific and predicts the outcome of a particular study	literature review; closing
Variables are quantitative or categorical; a sample is a subset of a population (convenience or simple random sampling)	method: participants, materials
Experimental research manipulates an independent variable; non-experimental research measures variables as they naturally occur	method: design

Key idea	Where it appears
Laboratory experiments: high internal validity; field studies: often more external validity	design; limitations
Descriptive statistics (mean, median, mode; range, standard deviation, variance) and inferential statistics (statistical significance)	planned results
Type I and Type II errors: we never know for certain that a conclusion is correct	limitations
Theories can be supported but not proved, and a disconfirmed hypothesis does not necessarily disprove a theory	discussion
The three levels of APA style and the standard sections of a report (Lessons 33 to 35)	the whole proposal

> **APA 7th edition: the order of a complete student paper**
>
> 1. Title page: bold title, your name, department and university, course, instructor, date.
> 2. Abstract: a new page, "Abstract" bold and centered, one paragraph with no indent, 250 words at most.
> 3. Text: a new page headed by the bold, centered title; introduction, then Method, Results and Discussion, which do not start new pages.
> 4. References: a new page.
> 5. Tables, then figures (each on its own page, unless placed in the text), then appendices, such as Appendix A, your consent form, and Appendix B, your questionnaire.
>
> Throughout: double spacing, 1-inch margins, paragraphs indented 0.5 inch, a page number at the top right, and a readable font such as 12-point Times New Roman or 11-point Calibri.

Language focus

1. Defining and contrasting. The summaries define with *be* or a simple present verb, add examples in parentheses, and contrast with *while*:

- "A theory is broad in nature and explains larger bodies of data. A hypothesis is more specific and makes a prediction about the outcome of a particular study."
- "Variables vary across people or situations and may be quantitative (e.g., age) or categorical (e.g., course subject)."
- "Laboratory experiments tend to have high internal validity …, while field studies often have more external validity …"
- *e.g.* gives examples; *i.e.* and *that is* restate: "measuring variables as they naturally occur (i.e., without manipulating anything)".

2. Careful claims about evidence. The summaries never claim certainty: "unlikely to be due to chance alone and therefore … likely to represent a real effect"; "we never know if our conclusions are correct"; theories can be supported but not proved; "disconfirming a hypothesis does not necessarily mean that theory has been disproved". Use the same care in your discussion: *support*, not *prove*; *suggest*, *is consistent with*.

> **Note for Arabic speakers**
>
> - Arabic nominal sentences need no verb "to be" in the present, so a definition can lose its verb in English: "A hypothesis more specific than a theory." Check that every definition has *is* or *are*.

Practice

1. Reading questions

1. Describe the cyclical model of research in one sentence.
2. Give four reasons for reviewing the literature early in a project.
3. Which factors affect the interestingness of a research question, and which affect its feasibility?
4. What do psychologists do in the hypothetico-deductive method?
5. Why do laboratory experiments and field studies differ in validity?
6. What is the difference between a Type I and a Type II error?
7. What does the method section contain at a minimum?
8. What is the function of talks and posters?

2. Book exercises (from 2.9)

1. "Practice: Describe why theories can be supported but not proved."
2. "Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice." Do the same for your own proposal.
3. "Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables." Answer the same question for your proposal.
4. "Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported."

3. Language exercise: proofread the book. Each item from the book's summaries contains at least one error. Correct it.

1. "Theories can be supported by not proved." (2.9)
2. "Since statistics are probabilistic in nature we never know if our conclusions are correct." (2.9)
3. Zajonc, R.B. & Sales, S.M. (1966). Social facilitation of dominant and subordinate responses. *Journal of Experimental Social Psychology, 2*, 160-168. (2.9)
4. The order of three references in 2.9: Zajonc, Heingartner, and Herman (1969); Zajonc (1965); Zajonc and Sales (1966).
5. American Psychological Association. (2010). *Publication Manual of the American Psychological Association* (6th ed.). Washington, D.C.: American Psychological Association. (11.5, placed last in the list, after Onwuegbuzie)

6. Adair, J. G., & Vohra, N. (2003). The explosion of knowledge, references, and citations: Psychology's unique response to a crisis. *American Psychologist, 58*, 15–23. (2.9; compare 11.2)

Speak with your tutor

- "Act as a professor reviewing my proposal. Ask me five hard questions about my hypothesis, design, measures and ethics, and I will defend my choices."
- "Let me explain why theories can be supported but not proved. Correct my reasoning and my English."
- "Here is my abstract: [paste]. Does it give the question, the method, the expected results and the main implication in 250 words or fewer?"
- "Here is one section of my final proposal: [paste]. Give me the three most important corrections, one for each level of APA style."

Writing task

Your research project: the complete proposal. Put together everything you wrote in this stage on sleep and memory in university students (or your own topic) as one APA-style student paper of about 2,000 to 3,000 words, plus references and appendices.

1. Assemble the parts in the order shown in the APA box: title page and introduction (Lesson 34), method, planned results and discussion (Lesson 35), references (Lesson 33), and appendices with your consent form (Lesson 09) and your measures (Lesson 12).
2. Write the abstract last (150 to 250 words): the research question, the planned method, the expected results and the main implication.
3. Make the parts consistent: the same title, variable names and operational definitions everywhere; a hypothesis that matches the planned analysis; the same design and invented numbers as in Lessons 27 to 36 (20 students, nap or wake, 40 word pairs), labelled; the future tense for the plan.
4. Revise with your tutor's earlier feedback, proofread with the checklist, and read the paper aloud once.

Self-editing checklist:

- Organization: all sections in APA order; the introduction has an opening, a literature review and a closing.
- The literature review is a balanced argument, and the closing states the hypothesis in operational terms.
- Another researcher could run the study from my method.
- High-level style: formal, straightforward and bias-free; the plan in the future tense; conclusions hedged (*support*, not *prove*).
- Citations: author and year, *et al.* for three or more authors, "&" only inside parentheses in the text; every citation matches a reference.
- References: alphabetical, hanging indents, sentence-case titles, DOIs as https://doi.org/ URLs.

- Format: double spacing, page numbers, bold headings (level 1 centered, level 2 flush left).
- Low-level errors from Lesson 33: numerals for 10 and above, hyphens, *since*, serial commas, abbreviations defined first.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- TED-Ed, "Not all scientific studies are created equal" (David H. Schwartz): https://youtu.be/GUpd2HJHUt8
- UBC iSchool, "How To Read an Academic Paper": https://youtu.be/SKxm2HF_-ko

Answer key

1. Reading questions

1. A research question based on the research literature leads to an empirical study, whose results are published and become part of the literature.
2. To refine the research question, identify appropriate methods, place the question in the context of other research, and prepare to write an effective report.
3. Interestingness: whether the answer is in doubt, whether it fills a gap in the literature, whether it has important practical implications. Feasibility: time, money, technical knowledge and skill, and access to special equipment and participants.
4. They construct theories (or work with existing ones), derive hypotheses from them, test the hypotheses, and reevaluate the theories in light of the results.
5. Laboratory experiments allow strong causal conclusions (high internal validity); field studies, in real settings, generalize better to the real world (more external validity).
6. Type I: concluding that an effect is real when it is not. Type II: concluding there is no effect when there is a real effect in the population.
7. A participants subsection and a design and procedure subsection.
8. To present new research to interested researchers and to facilitate further interaction among them; they are much less detailed than reports.

2. Book exercises

1. (example answer) A theory makes general claims, but each study tests one hypothesis derived from it, with one sample, setting and set of measures. A future study may still produce results that do not fit, and because statistics are probabilistic, any single result could be a Type I error. So evidence can support a theory but never prove it. Likewise, a disconfirmed hypothesis does not necessarily disprove the theory: the method, the measures or a Type II error may be responsible.
2. (example answer, for the default proposal) Research literature: studies on sleep and memory consolidation, found in PsycINFO or Google Scholar. Research question: do students who nap after learning remember more than students who stay awake? Empirical study: 20 students learn 40 word pairs, nap or rest awake for the same period, then recall the pairs. Publication: a poster at a student conference, which adds the results to the literature.
3. (example answer) The default proposal is experimental: students are randomly assigned to a nap or an equal period awake (the independent variable), and the number of word pairs recalled is the dependent variable. A study of students' usual hours of sleep and their memory scores would be non-experimental (correlational).
4. (example answer) Usually measures of central tendency (means, sometimes medians) and of variability (standard deviations, ranges), often in a table, plus percentages for categorical variables such as gender.

3. Language exercise

1. "Theories can be supported but not proved."
2. "Because statistics are probabilistic in nature, we never know whether our conclusions are correct." (*Since* only for time, in APA style; a comma after the introductory clause; optionally, the more formal *whether* for *if*.)
3. Zajonc, R. B., & Sales, S. M. (1966). Social facilitation of dominant and subordinate responses. *Journal of Experimental Social Psychology, 2*, 160–168. (Spaces between initials; a comma before "&"; an en dash in the page range.)
4. Zajonc (1965); Zajonc, Heingartner, and Herman (1969); Zajonc and Sales (1966). A one-author entry comes first; then the second author's name decides (H before S). The Zajonc (1965) entry also lacks its final period.
5. American Psychological Association. (2010). *Publication manual of the American Psychological Association* (6th ed.). Washington, DC: Author. It belongs at the top of the list. (Sentence case for book titles; "DC" as in the book's own Bem reference; the 6th edition wrote "Author" when the publisher was the author.) 7th edition: American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). https://doi.org/10.1037/0000165-000
6. Add the issue number and the DOI given in 11.2: … *American Psychologist, 58*(1), 15–23. https://doi.org/10.1037/0003-066X.58.1.15

My notes